business - an approximate algorithm, leveraging a random sharding with parallelization and then choosing the best of the solutions returned for each shard.

- If you are using an approximated algorithm, then usually you have a trade-off between performance and accuracy: in that case, you need to make sure you can compromise on the quality of the results, to obtain an answer within the time you can afford to wait.

11.3.2 Keeping the Inventory in Sync

If the situation doesn’t already look complicated enough to you, there is another troubling issue that we haven’t considered yet: where do we get the information about the availability of items?

So far, we have assumed this info is in the node of our k-d tree (or s-tree), but that might as well not be the case: if you think about it, when we place an order to a retailer, their inventory goes down, but the inventory in our container does reflect that.

There are a number of issues to take into account: the retailer could sell goods through other channels (either another e-commerce site, or a brick-and-mortar shop), and we need to update our copy of the inventory, when that happens; we need to communicate over a network, and so we need to be careful about race conditions to avoid placing orders twice or missing them.